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(54) **VEHICLE HAVING DRONE AS AN OFF
ROAD ASSISTANT**

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2201/10 (2023.01); *B64U 2201/20* (2023.01)

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B64U 2101/30; *G06V 20/588*; *G08G*
5/74; *G08G 5/76*

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See application file for complete search history.

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(57) **ABSTRACT**

A system and method including an aerial drone associated with a vehicle that can serve as a surveillance tool for previewing an off-road trail that may be travelled by the vehicle. The vehicle may include a vehicle electronic control unit and drone docking station that are configured to communicate with and control the drone during surveillance flight operations.

12 Claims, 8 Drawing Sheets

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(51) **Int. Cl.**

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B64U 101/30 (2023.01)

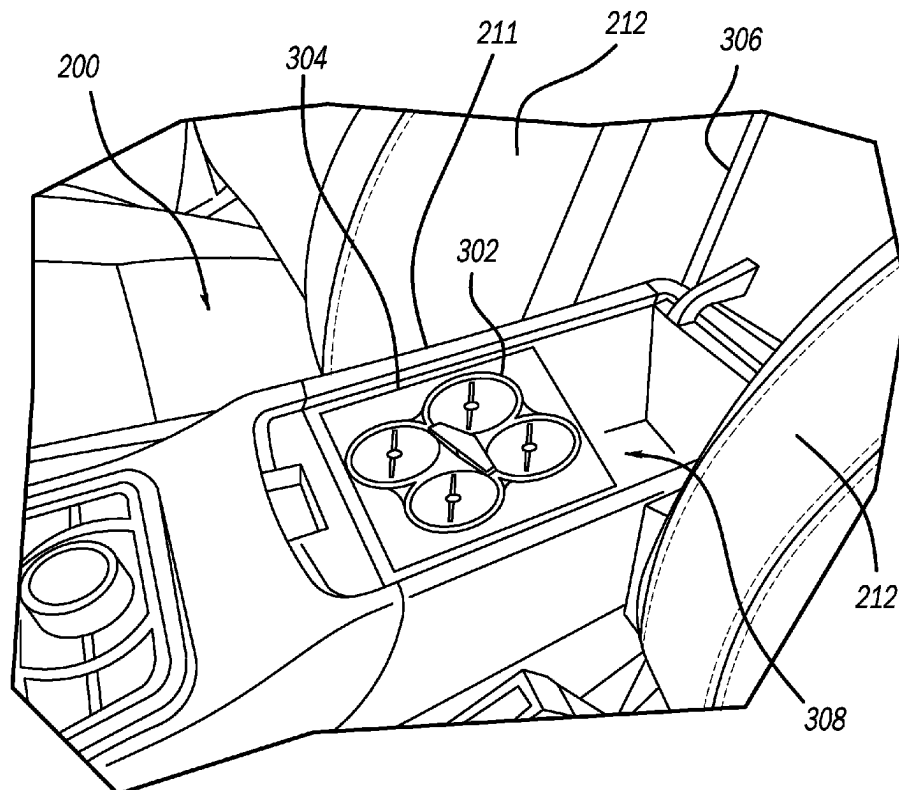
G06V 20/56 (2022.01)

G08G 5/74 (2025.01)

G08G 5/76 (2025.01)

(52) **U.S. Cl.**

CPC *B64U 80/86* (2023.01); *G06V 20/588*
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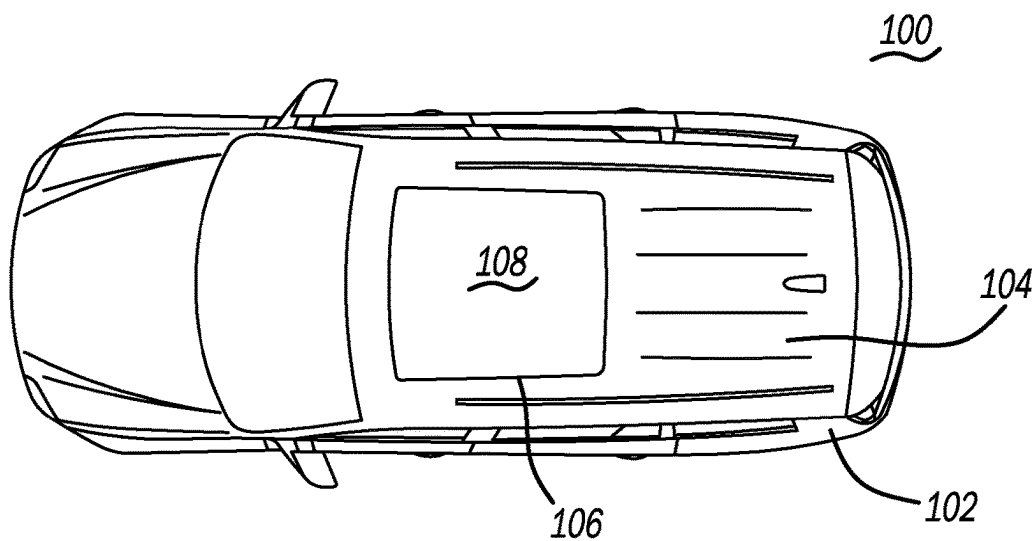


FIG. 1

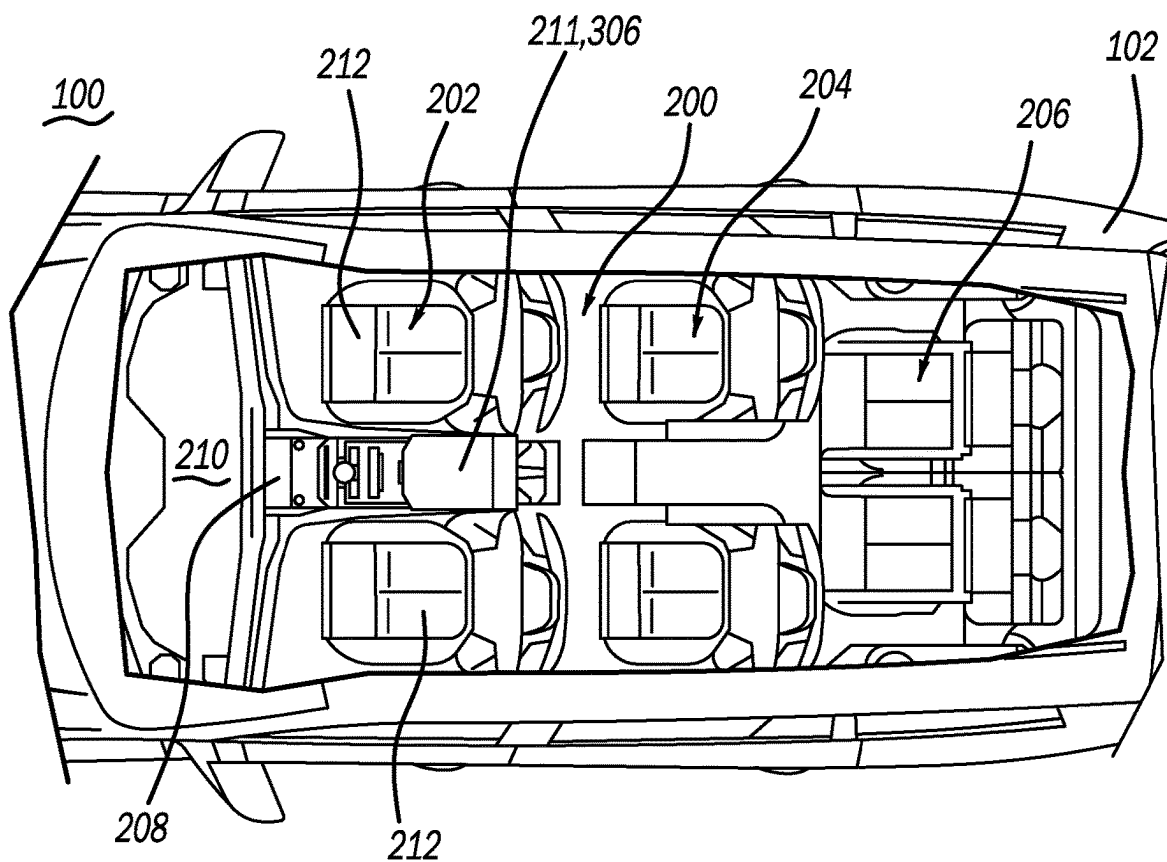


FIG. 2

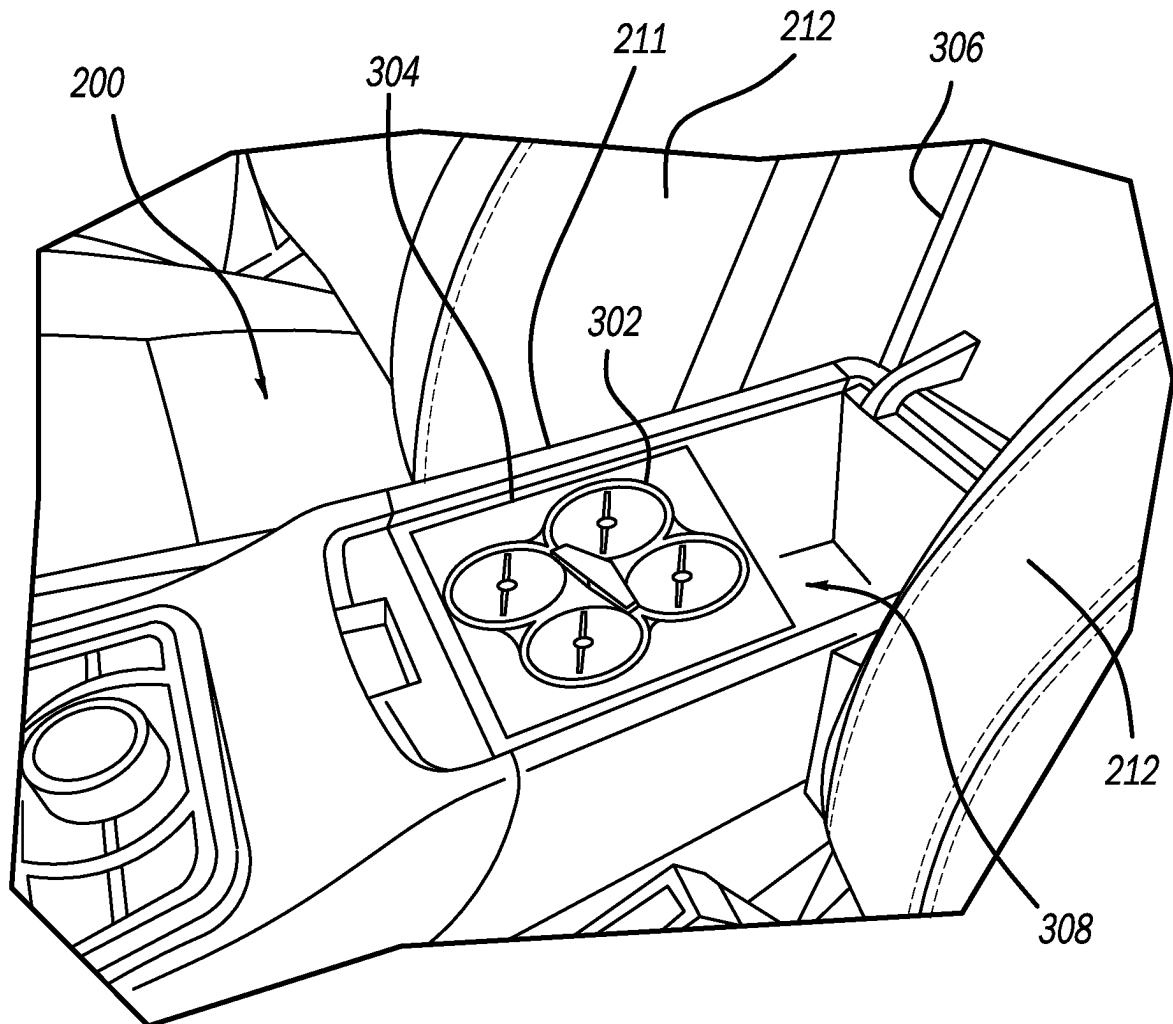


FIG. 3

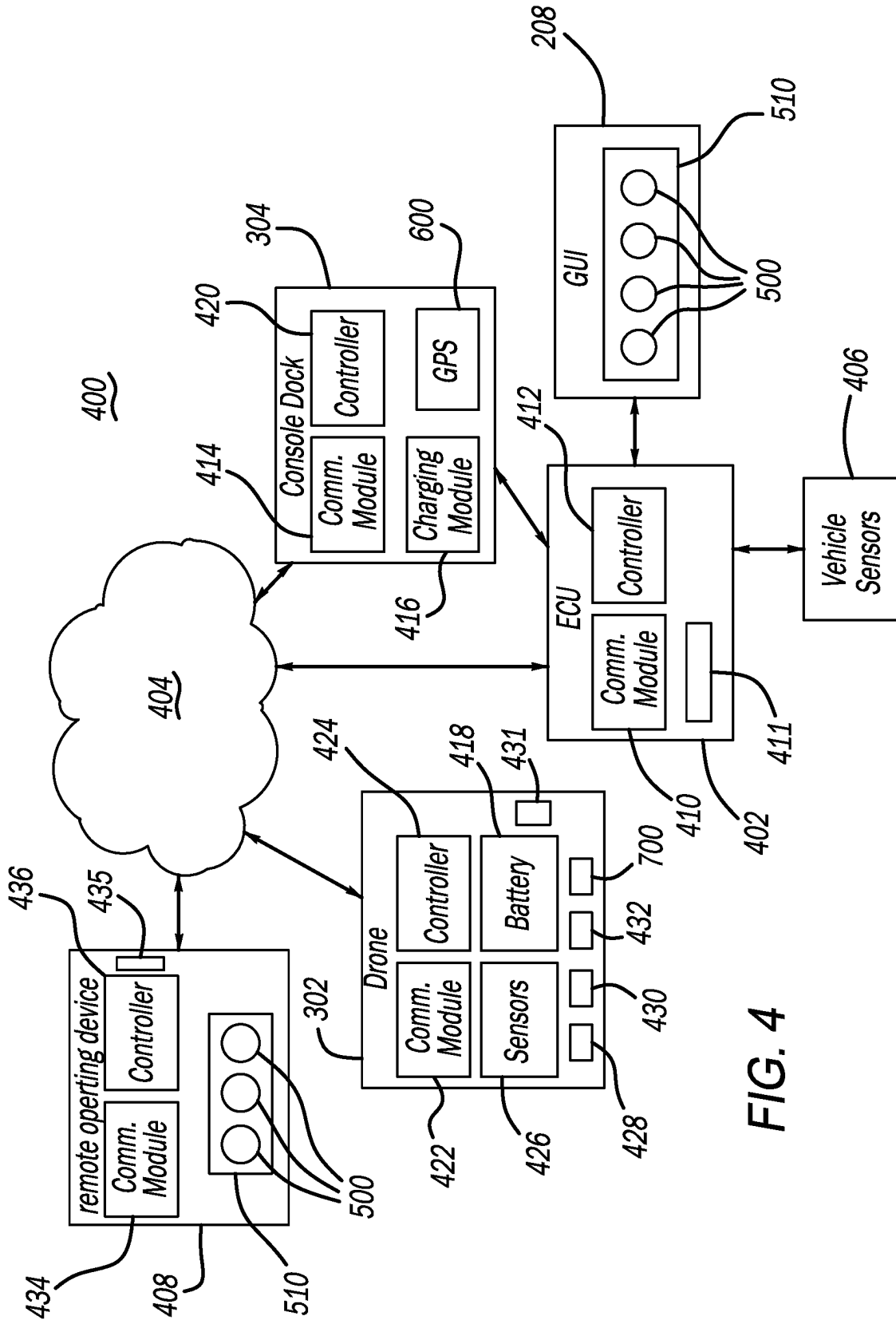


FIG. 4

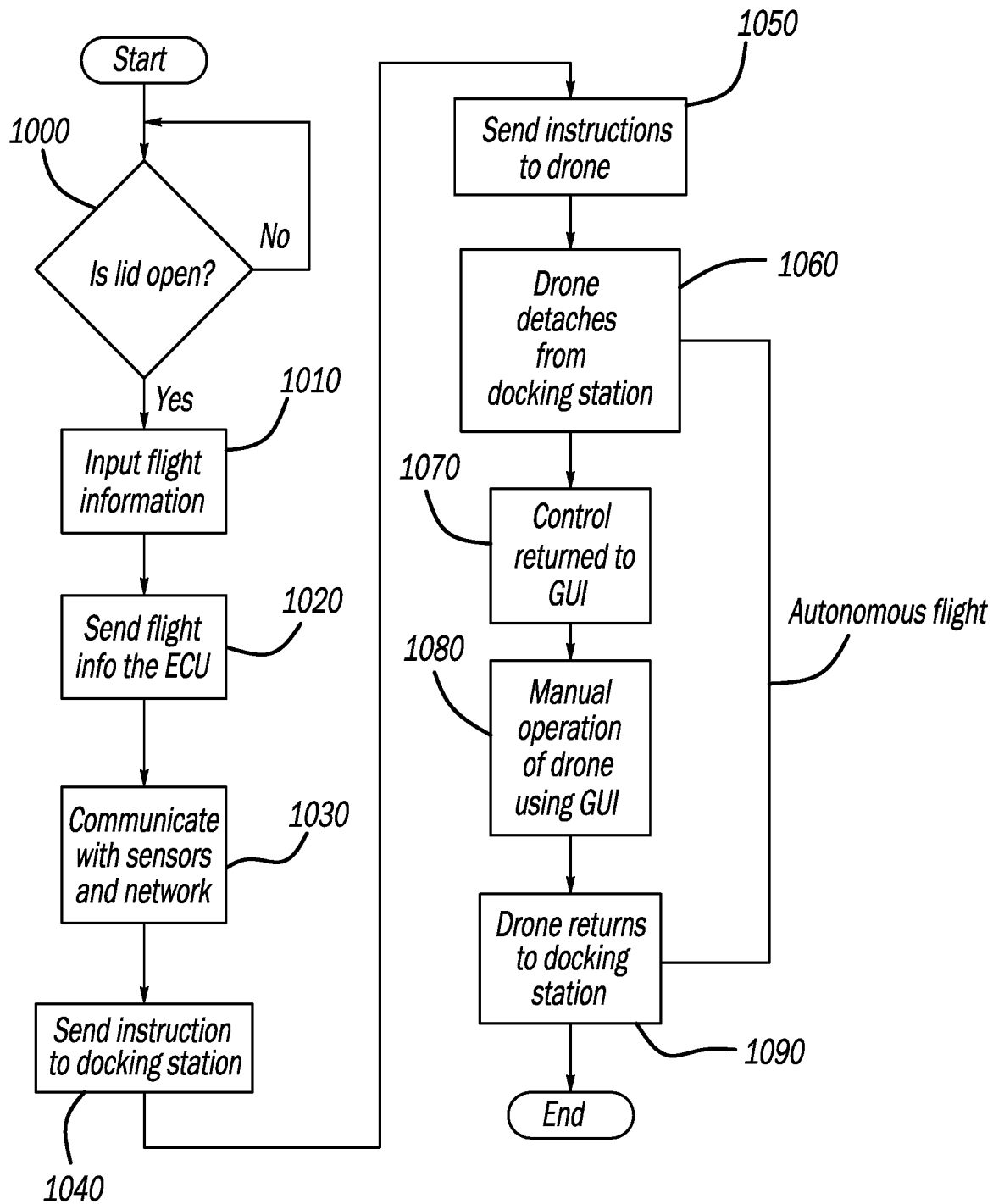


FIG. 5

208,408

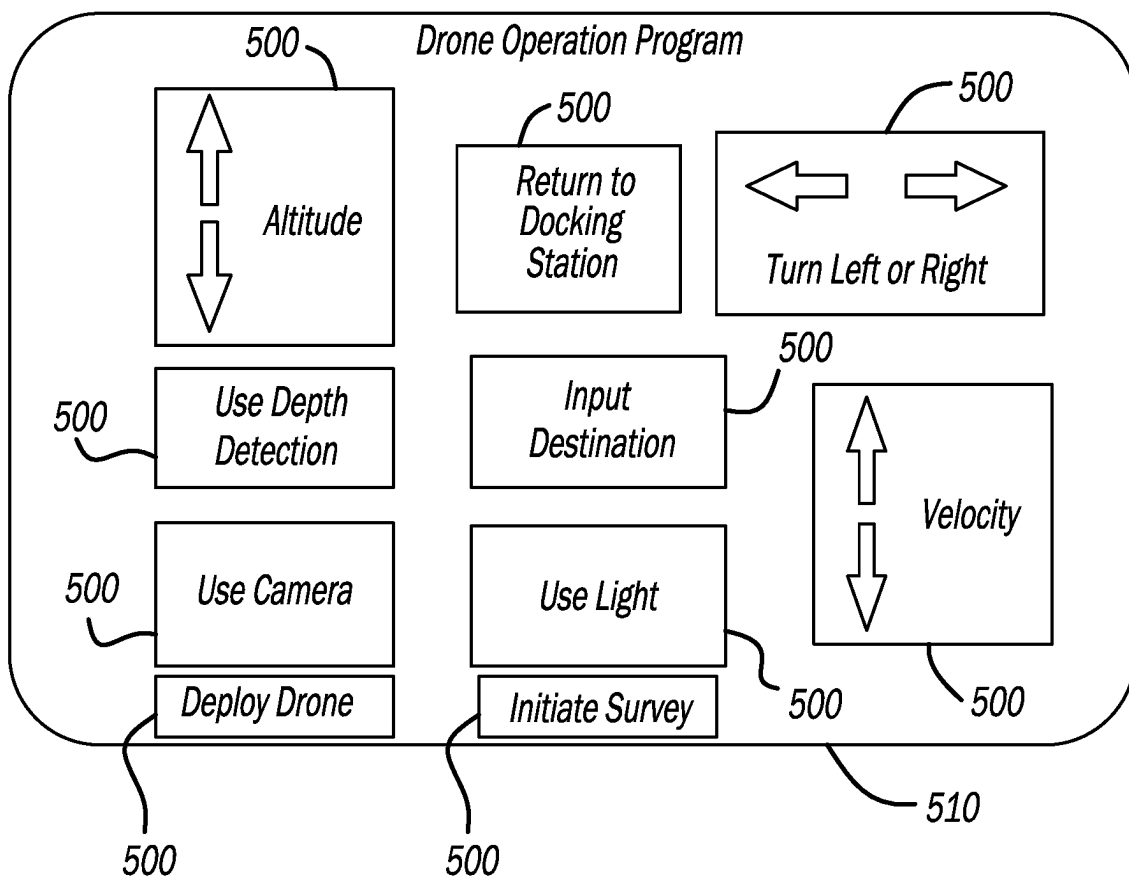


FIG. 6

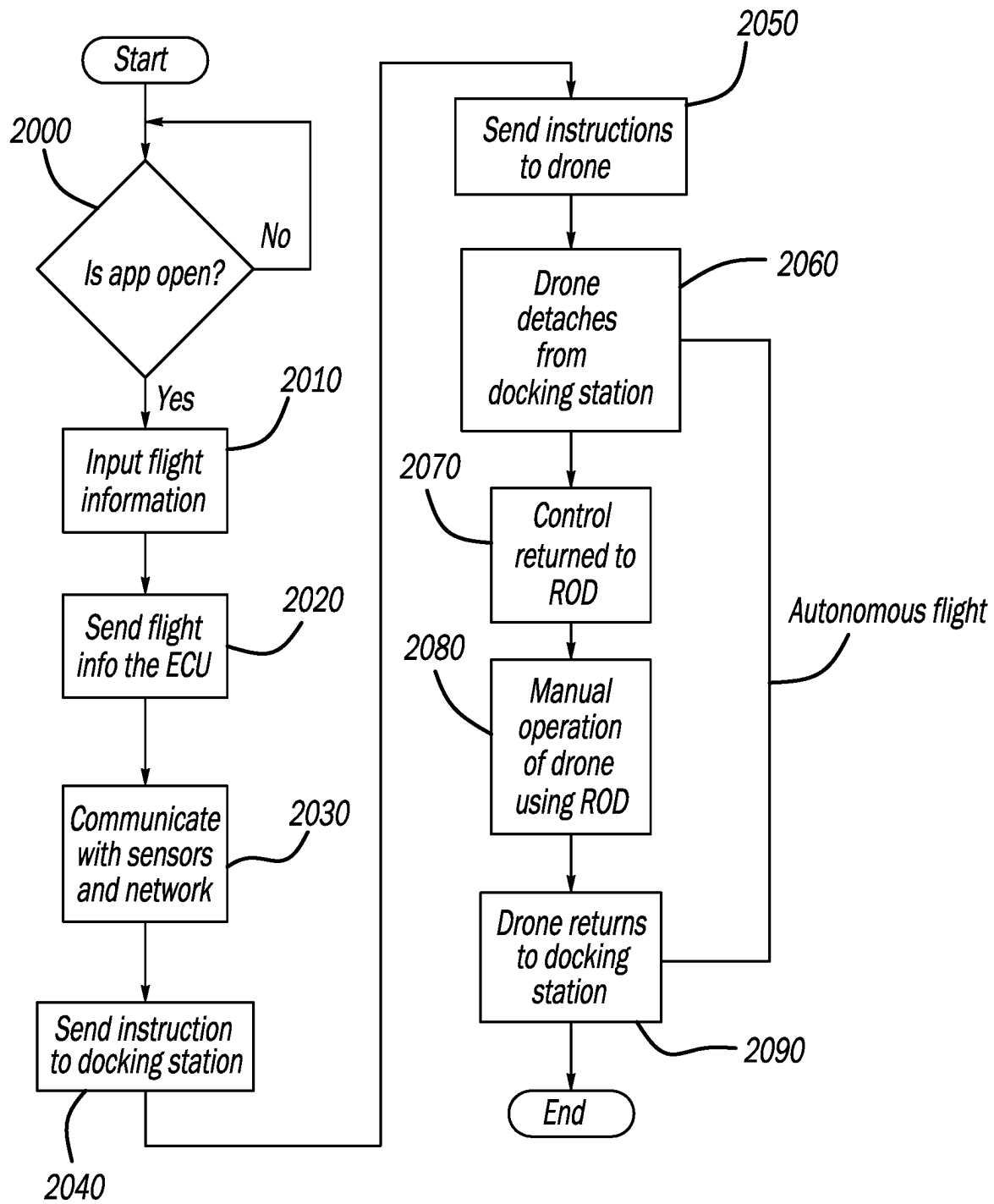


FIG. 7

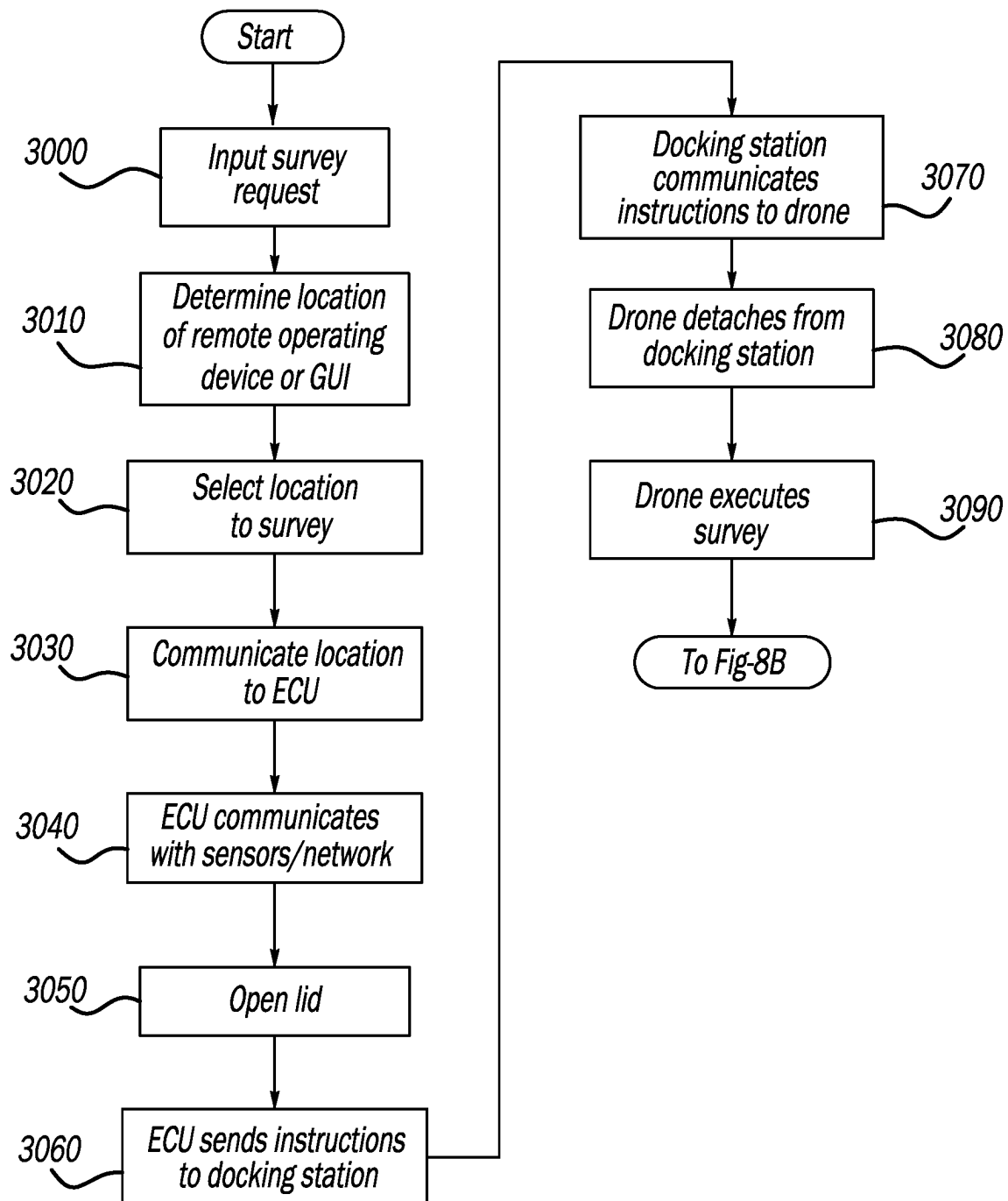
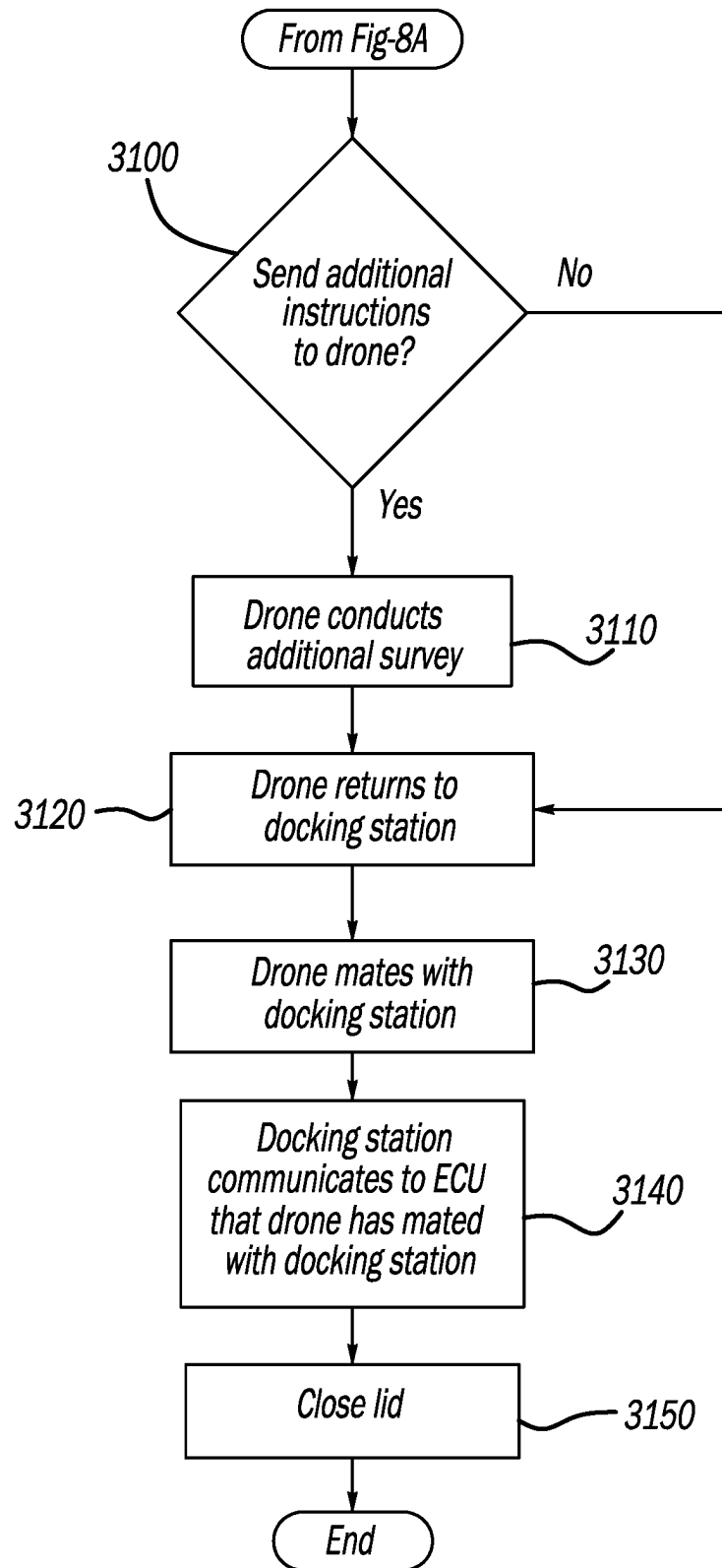


FIG. 8A

**FIG. 8B**

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VEHICLE HAVING DRONE AS AN OFF ROAD ASSISTANT

FIELD

The present disclosure relates to a vehicle having a drone as an off-road assistant.

BACKGROUND

This section provides background information related to the present disclosure which is not necessarily prior art.

Many people enjoy using vehicles that are equipped to travel off-road. Indeed, there are many off-road trails throughout the United States and other countries that have various levels of difficulty. For example, some off-road trails have large rocks and bodies of water that must be traversed while in the vehicle. Typically, before various off-road trails that have various obstacles (e.g., large rocks and/or bodies of water) are driven using the vehicle, an occupant of the vehicle may first walk the off-road trail to determine whether any substantial obstacles are located along the off-road trail, which can be time-consuming. It is desirable, therefore, to provide a way to avoid having to first walk the off-road trail to determine the location of obstacles before using the vehicle to travel along the off-road trail.

SUMMARY

This section provides a general summary of the disclosure and is not a comprehensive disclosure of its full scope or all of its features.

According to a first aspect of the present disclosure, there is provided a system for controlling an aerial drone, including a vehicle body defining an interior; a vehicle electronic control unit (ECU) located within the interior of the vehicle; a drone docking station positioned within the interior of the vehicle and configured to mate with the aerial drone, and including a docking station controller configured to communicate with the vehicle ECU; and at least one of a graphic user interface (GUI) provided in the interior of the vehicle and configured to communicate with the vehicle ECU and a remote operating device configured to communicate with the vehicle ECU, each of the GUI and remote operating device being configured for receipt of an input to conduct a surveillance flight using the aerial drone, wherein upon input of a request to either the GUI or remote operating device to conduct the surveillance flight using the drone, the GUI or remote operating device is configured to transmit a signal indicative of the request to the vehicle ECU, upon receipt of the request by the ECU, the ECU is configured to communicate an instruction to the drone docking station controller to initiate surveillance flight operations using the aerial drone, upon receipt of the instruction by the drone docking station controller to initiate surveillance flight operations, the drone docking station controller communicates an instruction to the aerial drone to initiate flight and conduct the surveillance flight operation.

According to the first aspect, the surveillance flight operation surveys an off-road trail that is to be traversed using the vehicle.

According to the first aspect, the system may include a network in communication with at least one of the vehicle ECU and drone docking station controller, wherein at least one of the vehicle ECU and drone docking station controller is configured to communicate with the aerial drone via the network.

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According to the first aspect, during the surveillance flight operations, the aerial drone is configured to generate at least one of images and video while in flight and transmit the images and video to at least one of the vehicle ECU and drone docking station via the network, and upon receipt of the images and video from the network, at least one of the vehicle ECU and drone docking station are configured to transmit the images and video to the GUI or remote operating device for display thereon.

According to the first aspect, during the surveillance flight operations, the aerial drone is configured to generate at least one of images and video while in flight and save the images and video in a memory, and after conclusion of the surveillance flight operations by the aerial drone, and the aerial drone is mated with the drone docking station, the drone docking station is configured to obtain the images and video from the memory of the aerial drone and communicate the images and video to the vehicle ECU, which then communicates the images and video to either the GUI or remote operating device for display thereon.

According to the first aspect, before communicating the instruction to the drone docking station controller to initiate surveillance flight operations using the aerial drone, the vehicle ECU is configured to communicate with at least one of the network and vehicle sensors to obtain information indicative of weather conditions that can impact the surveillance flight operation.

According to the first aspect, based on the information indicative of the weather conditions, the vehicle ECU is configured to communicate instructions to the drone docking station that either permit the surveillance flight operations to occur or prevent the surveillance flight operations from occurring.

According to a second aspect of the present disclosure, there is provided a method of using an aerial drone to conduct surveillance of an off-road trail before travelling the off-road trail using an off-road vehicle, including receiving, by either a remote operating device or a graphic user interface (GUI) of the off-road vehicle, a request to conduct a surveillance flight using the aerial drone; displaying on a screen of either the remote operating device or GUI a map that displays the off-road trail; selecting a location of interest along the off-road trail that is transmitted by either the remote operating device or GUI to a vehicle electronic control unit (ECU), the vehicle ECU being configured to communicate with the aerial drone; generating, with the vehicle ECU, an instruction to the aerial drone to conduct surveillance of the location of interest along the off-road trail.

According to the second aspect, the off-road vehicle includes an interior that includes a drone docking station that is configured to mate with the aerial drone, the drone docking station including a drone docking station controller that communicates with each of the aerial drone and the vehicle ECU, and the method further comprises transmitting the instruction from the vehicle ECU to the drone docking station controller.

According to the second aspect, the method may further include communicating from the drone docking station to the aerial drone the instruction generated by the vehicle ECU.

According to the second aspect, the off-road vehicle includes a plurality of sensors in communication with the vehicle ECU that are configured to generate signals indicative of weather conditions being experienced by the off-road vehicle, and before generating the instruction to conduct surveillance, the vehicle ECU is configured to determine,

based on the signals indicative of the weather received from the plurality of sensors, whether the weather conditions are satisfactory for conducting the surveillance of the off-road trail.

According to the second aspect, the vehicle ECU is configured to communicate with a network and obtain information indicative of the weather conditions from the network, and before generating the instruction to conduct surveillance, the vehicle ECU is configured to determine, based on the information indicative of the weather received from the network, whether the weather conditions are satisfactory for conducting the surveillance of the off-road trail.

According to the second aspect, the vehicle ECU and aerial drone are each in communication with a network, and during the surveillance of the off-road trail using the aerial drone, the aerial drone is configured to transmit images of the off-road trail to the vehicle ECU via the network, and wherein the method further comprises transmitting the images received by the vehicle ECU to either the remote operating device or GUI for viewing on the screen.

Further areas of applicability will become apparent from the description provided herein. The description and specific examples in this summary are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

DRAWINGS

The drawings described herein are for illustrative purposes only of selected embodiments and not all possible implementations, and are not intended to limit the scope of the present disclosure.

FIG. 1 is an overhead view of a vehicle according to a principle of the present disclosure;

FIG. 2 is an overhead view of an interior of the vehicle illustrated in FIG. 1;

FIG. 3 is a perspective view of an example drone docking station having a drone mounted thereto, located within the interior of the vehicle illustrated in FIG. 2;

FIG. 4 is a schematic diagram depicting a vehicle system including a drone docking system according to a principle of the present disclosure;

FIG. 5 is a flowchart depicting an example method for controlling a drone using the drone docking system of the present disclosure;

FIG. 6 is a flowchart depicting another example method for controlling a drone using the drone docking system of the present disclosure;

FIG. 7 is a flowchart depicting another example method for controlling a drone using the drone docking system of the present disclosure; and

FIGS. 8A and 8B are flowcharts depicting an example method of using a drone to pre-survey a travel path of the vehicle illustrated in FIG. 1.

Corresponding reference numerals indicate corresponding parts throughout the several views of the drawings.

DETAILED DESCRIPTION

Example embodiments are provided so that this disclosure will be thorough, and will fully convey the scope to those who are skilled in the art. Numerous specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of embodiments of the present disclosure. It will be apparent to those skilled in the art that specific details need not be employed, that example embodiments may be embodied in many

different forms and that neither should be construed to limit the scope of the disclosure. In some example embodiments, well-known processes, well-known device structures, and well-known technologies are not described in detail.

FIG. 1 illustrates an example vehicle **100** that may include a drone **302** and drone docking station **304** (FIG. 3) according to the present disclosure. Vehicle **100** has a body **102** defining a roof **104**. In the illustrated embodiment, roof **104** may define an opening **106**. Opening **106** may include, for example, a glass panel **108** defining a sun roof, a moon roof, or a panoramic roof that permits light to enter the vehicle. Panel **108** does not necessarily need to be formed of glass or a transparent material. In contrast, panel **108** may be an opaque removable roof panel. In either case, it should be understood that panel **108** is movable to provide access to an interior **200** (FIG. 2) of vehicle **100**.

FIG. 2 illustrates an example interior **200** of vehicle **100**. In the illustrated example, interior **200** includes a front row of seats **202**, a second row of seats **204**, and a third row of seats **206**. It should be understood, however, that second and third rows of seats **204** and **206** may be omitted such that vehicle includes a rear cargo storage area in place thereof. Interior **200** may also include a graphic user interface (GUI) **208** on a dashboard **210** of vehicle **100** that is in communication with an electronic control unit **402** (FIG. 4) of vehicle **100**. Interior **200** may include a cargo bin **211** located proximate GUI **208** between the seats **212** of first row **202**.

As best shown in FIG. 3, cargo bin **211** may have a movable lid **306** that may be opened to expose and provide access to a storage area **308** of cargo bin **202**, and closed to enclose and prevent access to storage area **308**, as is known in the art. In accordance with the present disclosure, storage area **308** may include a docking station **304** configured to mate with a drone **302**. As will be described in more detail later, lid **306** may be opened to permit drone **302** to detach from docking station **304** and exit the storage area **308** while being controlled by an operator. Alternatively, the operator may simply open lid **306**, manually detach drone **302** from docking station **304**, manually remove drone **302** from vehicle **100**, and then operate drone **302**. If operator elects to control drone **302** remotely to exit storage area **308** and vehicle **100**, the operator may first open or remove panel **108** located on roof **104** of vehicle **100** to permit drone **302** to exit vehicle **100** via flight.

Now referring to FIG. 4, a schematic representation of a vehicle system **400** for controlling drone **302** that includes drone docking station **304** is illustrated. In addition to docking station **304**, system **400** may include ECU **402**, GUI **208**, a communication network **404**, various vehicle sensors **406**, and an optional remote operating device **408**. Example remote operating devices **408** include remote controllers, mobile phones, portable or desktop computers, electronic tablets, and the like.

ECU **402** includes a communication module **410** for communicating with various modules located throughout vehicle **100** that control various features of the vehicle such as, for example, climate control modules (not shown), vehicle exterior and interior lighting modules (not shown), engine control modules (not shown), window control modules (not shown), and the like. Communication module **410** also enables ECU **402** to communicate with GUI **208**, docking station **304**, and network **404**. ECU **402** also includes a memory **411** such as a RAM or ROM, and a controller **412** that communicates with communication module **410**, wherein controller **412** is configured to communicate various instructions to GUI **208** and docking station **304**.

(as well as other vehicle modules like those described above) via communication module 410.

ECU 402 is also in communication with vehicle sensors 406. Example vehicle sensors 406 that may communicate with ECU 402 include interior and exterior temperature sensors (not shown), rain sensors (not shown), exhaust gas sensors (not shown), engine load sensors (not shown), oxygen sensors (not shown), fuel level sensors (not shown), and other vehicle sensors known to one skilled in the art. Specific sensors 406 that may communicate with ECU 402 to assist with operating drone 302 include rain sensors that are configured to transmit signals indicative of whether vehicle 100 is being exposed to rain, velocity sensors that are configured to transmit signals indicative of a velocity of the vehicle 100, wiper motor sensors that are configured to transmit signals indicative of the vehicle windshield wiper blades (not shown) being operated, and accelerometer or gyro-sensors that are configured to transmit signals indicative of an acceleration of the vehicle 100 or whether the vehicle 100 is located on a slope (i.e., not located on flat ground). These sensors 406 are important because these sensors 406 can generate signals that assist in determining whether to deploy or recall drone 302, or assist in operating drone 302 when it enters or exits vehicle 100.

For example, rain sensors can communicate signals indicative of whether it is raining. Operation of drone 302 may not be desirable in adverse weather conditions. Thus, based on a signal indicative of whether it is raining transmitted by the sensor 406, ECU 402 may be configured to send an instruction to docking station 304 that either prevents drone 302 from being deployed in the adverse weather conditions, or send an instruction to docking station 304 to recall a deployed drone 302 back to the docking station 304 so as to avoid drone 302 from being operated and/or damaged by the adverse weather conditions. Windshield wiper motor sensors 406 may be used in the same manner. These operations will be described in more detail later.

It also may not be desirable to deploy drone 302 from vehicle 100, or recall drone 302 back to vehicle 100, when vehicle 100 is in motion. Accordingly, signals indicative of the velocity of vehicle 100 or an acceleration of vehicle 100 that are transmitted to ECU 402 can be used to assist in operation of drone 302. For example, if ECU 402 receives a signal indicative of the velocity or acceleration of the vehicle 100, ECU 402 may communicate an instruction to docking station 304 that prevents drone 302 from being deployed from docking station 304 while the vehicle 100 is in motion. Conversely, if drone 302 is already deployed outside of vehicle 100 and the vehicle 100 is in motion, ECU 402 can send an instruction to docking station 304, which is subsequently communicated to drone 302, that prevents drone 302 from returning to docking station 304. In this regard, it is much more difficult to return drone 302 to the docking station 304 when the vehicle 100 is in motion and such an instruction can prevent drone 302 from attempting to dock with docking station 304 while the vehicle 100 is in motion to prevent unnecessary damage to either vehicle 100 or drone 302 in such an instance.

Docking station 304 includes a docking station communication module 414 that communicates with ECU 402 and network 404, a charging module 416 that is configured to charge a battery 418 of drone 302 when drone 302 is mated with docking station 304, a docking station controller 420 that is configured to send instructions to drone 302 (via network 400 when drone 302 is deployed, or directly to drone 302 via communication module 414 when drone 302 is mated with docking station 304) and to charging module

416, and a GPS module 600. Docking station controller 420 may send instructions to charging module 416 to begin charging drone 302 when drone 302 is mated with docking station 304, or send instructions to charging module 416 to cease charging drone 302 upon receive of a signal indicative that battery 418 of drone 302 has fully charged.

GUI 208 communicates with ECU 402 and may be a touch-screen operated GUI 208. As is known in the art, GUI 208 can be used to operate various systems of vehicle 100. For example, an occupant of vehicle 100 can operate a climate control system or a radio of the vehicle 100 using GUI 208 by touching various icons 500 on a screen 510 of the GUI 208. Drone 302 may be operated from GUI 208 in a similar manner. For example, if an occupant of vehicle 100 elects to deploy drone 302 from the docking station 304, GUI 208 may include various icons 500 that may be touched to operate drone 30 as best shown in FIG. 6. For example, a “deploy drone” icon 500 may be located on GUI 208 and upon the icon 500 being touched by the occupant, a signal may be sent from GUI 208 to ECU 402, which then sends a signal to docking station 304 to deploy the drone 302.

GUI 208 may also be configured to display information relative to drone 302 such as, for example and without limitation, whether drone 302 is connected to docking station 304 or whether battery 418 of drone 302 is charged (e.g., by displaying a percentage of charge remaining in battery 418). If drone 302 is in use outside of vehicle 100, GUI 208 may also be used to control movement of drone 302 (e.g., an elevation of flight, velocity of drone 302, and the like). Any instruction input into GUI 208 is communicated to ECU 402, which is then communicated to docking station 304, which is then communicated to drone 302 via network 404. An example type of GUI 208 is a Uconnect® system.

Network 404 may be a wireless communication network operating in accordance with one or more standards including, but not limited to, Wi-Fi, 3GPP (3rd Generation Partnership Project), 4GPP (4th Generation Partnership Project), 5GPP (5th Generation Partnership Project), LTE (long term evolution), LTE Advanced, RFID, IEEE 802.11, Bluetooth, AMPS (advanced mobile phone services), digital AMPS, GSM (global system for mobile communications), CDMA (code division multiple access), LMDS (local multi-point distribution systems), MMDS (multi-channel-multi-point distribution systems), V2V (vehicle-to-vehicle), V2X (vehicle-to-everything), and/or variations thereof. In the illustrated embodiment, network 404 may independently communicate with drone 302, docking station 304, ECU 402, and remote operating device 408. Network 404 may also serve as a wireless conduit of communication between these devices.

Drone 302 includes a drone communication module 422, a drone controller 424, battery 418, various sensors 426, a memory 431 such as a RAM or ROM, and a GPS module 700. Optional components include a camera 428, a light-emitting device 430 that can be operated in a manner similar to a vehicle's headlights (e.g., to illuminate the ground beneath drone 302 during flight operations), and a depth device 432 configured to determine a depth of a puddle or other body of water. For example, depth device 432 can emit a laser, sonar, or some other type of energy that can assist in determining a depth).

Drone communication module 422 is configured to receive and transmit information from network 400 to other elements of the system such as, for example, docking station 304 and remote operating device 408. If drone 302 is mated

with docking station 304, drone communication module 422 may receive and transmit information directly to docking station control module 414.

Drone sensors 426 may be similar to vehicle sensors 406. Example drone sensors 426 include an altimeter that generates a signal indicative of an altitude of the drone 302, a GPS sensor that generates a signal indicative of a location of the drone 302 during flight operations, a velocity sensor that generates a signal indicative of a velocity of the drone 302 during flight operations, a rain sensor that generates a signal indicative of whether it is raining, a battery sensor that generates a signal indicative of a percentage of battery power remaining in battery 418, accelerometers that generate signals indicative of an acceleration of drone 302, and gyro-sensors that generate signals indicative of an orientation of drone 302. These signals may be transmitted to drone controller 424, which are then sent to communication module 422 to be transmitted to other elements of the system 400 via network 404.

If drone 302 includes a camera 428, images or video footage captured by camera 428 may be transmitted via communication module 422 to vehicle 100 or remote operating device 408. For example, any images or footage captured by camera 428 may be communicated to vehicle 100 (i.e., to ECU 402 via network 400, or to docking station 304 via network 400 and then to ECU 402 from the docking station 304) to be displayed by GUI 208 on screen 510. Alternatively, if remote operating device 408 includes a display 510, the images or footage captured by camera 428 transmitted to remote operating device 408 via network 404 may be displayed by remote operating device 408. In another alternative, the images and/or video obtained by camera 428 may be stored in memory 431 to be downloaded and viewed later.

Remote operating device 408 may include a remote operating device communication module 434, a GPS module 435, and a remote operating device controller 436. Remote operating device communication module 434 permits remote operating device 408 to communicate with network 404 and other elements of system 400 such as drone 302, docking station 304, and ECU 402. Remote operating device controller 436 may include a drone operating system software stored therein for operating drone 302. Based on inputs received from an operator, remote operating device controller 436 may then communicate instructions to any of drone 302, docking station 304, and ECU 402 via remote operating device communication module 434.

For example, if someone using remote operating device 408 desires to deploy drone 302 from docking station 304, the operator may use remote operating device 408 to send an instruction to docking station 304 via network 404, which then communicates with drone 302 to initiate deployment of drone 302 from docking station 304 (in such a case, it is assumed that panel 108 (FIG. 1) is open or removed).

In another embodiment, the instructions sent from remote operating device 408 to docking station 304 via network 404 may be further communicated to ECU 402. Upon receipt of such an instruction by ECU 402, ECU 402 may communicate with a vehicle sensor 406 such as, for example, a sensor 406 that generates a signal indicative of whether panel 108 is open or removed) to determine whether it is safe for drone 302 to exit the vehicle 100 through the opening 106 formed in roof 104 of vehicle 100. After ECU 402 receives a signal indicative of panel 108 being open or removed so as to not obstruct opening 106, ECU 402 may then communicate a signal back to docking station 304 that permits drone 302 to be deployed an exit vehicle 100 through opening 106.

In other embodiments, ECU 402 may also communicate with other sensors 406 or network 404 to determine whether weather conditions are such that drone 302 may be deployed. For example, if sensor 406 is a rain sensor that communicates a signal indicative of the vehicle 100 being exposed to rain to ECU 402, ECU 402 may transmit an instruction to docking station 304 that prevents drone 302 from being deployed. Alternatively, ECU 402 may communicate with network 404 to retrieve current weather conditions. Based on the information regarding the current weather conditions received from network 404, ECU 402 is configured to send an instruction to docking station 304 that either permits or prevents drone 302 from deploying from docking station 304.

Drone 302 may be any type of drone known to one skilled in the art, provided that drone 302 is sized to fit within vehicle 100. Examples of types of drones include, but are not limited to, a helicopter drone, a fixed-wing drone, a multi-rotary-wing drone, a blimp drone, a flexible-wing drone, and the like. Drone 302 may be capable of both autonomous navigation and operating via remote control.

Now referring to FIGS. 5 and 6, a method of operating drone 302 will be described. As noted above, drone 302 may be operated using GUI 208 or remote operating device 408. Description of using GUI 208 to operate drone 302 will first be described. Firstly, if docking station 304 and drone 302 are located within cargo bin 202, lid 306 must be opened (step 1000) by an occupant of vehicle 100 (this step may be omitted if, for example, docking station 304 is located in a rear storage area of vehicle 100 (i.e., if vehicle 100 does not include second or third rows 204, 206 of seats and instead includes a rear storage area)).

After opening lid 305, an occupant may then use GUI 208 to initiate flight operations of drone 302 (step 1010). At this time, an occupant of vehicle 100 may input information into GUI 208 such as, for example, a destination for drone 302 to travel, a desired flight path, a desired altitude for the flight, a time period of the flight, and the like by interacting with various icons 500 (FIG. 6) on a screen 510 of the GUI. At this time, GUI 208 communicates a signal to ECU 402 that drone 302 is to be deployed (step 1020) and that includes the desired information input to the GUI 208 by the occupant.

Upon receipt of this signal and information, ECU 402 communicates with various sensors 406 and/or network 404 to determine whether conditions are acceptable to initiate deployment of drone 302 (step 1030). These conditions include obtaining information relative to current weather conditions from network 404, or obtaining signals from sensors 406 that are indicative of the weather that vehicle 100 is currently being exposed (i.e., a signal indicative of rain from rain sensors or a signal indicative of the vehicle wiper blades being operated, for example). Other conditions include whether the vehicle 100 is in motion, whether the vehicle 100 is at an orientation (i.e., on a slope) that would make it difficult to deploy drone 302 from vehicle 100, and whether panel 108 is open or removed to permit drone 302 to exit the vehicle 100 during flight. If these conditions are satisfactory, ECU 402 may transmit an instruction (step 1040) to docking station 304 to permit drone 302 to begin flight operations, detach from docking station 304, and exit vehicle 100 through opening 106 provided in roof 104 of vehicle 100.

Upon receipt by docking station 304 of instructions from ECU 402, docking station 304 communicates instructions to drone 302 to initiate flight and including the desired flight characteristics input into GUI 208 by the occupant (step 1050). Upon receipt of these instructions, drone 302 may

autonomously detach from docking station 304 and exit vehicle 100 through opening 106 (step 1060). At this time, drone 302 may continue to operate autonomously using the desired flight characteristics input by the occupant to GUI 208 before returning to docking station 304.

Alternatively, drone 302 may autonomously detach from docking station 304 and exit vehicle 100 before operation of drone 302 is transferred back to GUI 208 (step 1070). In such a case, drone 302 may exit vehicle 100 and reach a predetermined altitude before operation of drone 302 is transferred back to GUI 208. Then, the occupant can use various control icons on GUI 208 to move in a particular direction, move at a desired velocity, reach a desired altitude, use camera 428, operate light source 430, operate depth detection device 432, and the like (step 1080). At this time, the instructions input into GUI 208 are communicated to ECU 402, which then communicates the instructions to docking station 304, which communicates the instructions to drone 302 via network 404.

If drone 302 is operating autonomously based on instructions input to GUI 208 and the desired flight time has elapsed, drone 302 may then automatically return to vehicle 100, pass through opening 106, and mate with docking station 304 (step 1090). Alternatively, if drone 302 is being manually operated, in step 1090 the occupant can input instructions into GUI 208 that direct drone 302 back to vehicle 100. The occupant can operate drone 302 to reenter vehicle 100 and mate with docking station 304, or the occupant can operate drone 302 to return to a location exterior to vehicle 100 before manually retrieving drone 302 and manually mating drone 302 with docking station 304. It should be understood that if vehicle 100 has moved during flight operations, a location of vehicle 100 can be transmitted to drone 302 via network 404. In this regard, both drone 302 and docking station 304 may include a GPS module 600, 700 that may coordinate with each other.

It should be understood that during flight operations, ECU 402 may continually communicate with network 404 and/or sensors 406 to continually make a determination whether conditions are satisfactory for continuing flight operations (e.g., continually checking weather information). If for whatever reason ECU 402 determines that flight operations should cease, ECU 402 may send an instruction to docking station 304 to have drone 302 return to vehicle 100, which is then communicated to drone 302 using network 404. Moreover, it should be understood that information obtained from drone sensors 426 can be continuously transmitted from drone 302 to docking station 304 and ECU 402 via network 404. If drone sensors 426 generate signals indicative of poor operation (e.g., motor issues, poor flight conditions, and the like), ECU 402 can transmit instructions to docking station 304 that are transmitted to drone 302 via network 404 to cease flight operations and return to vehicle 100.

A similar process can be used if drone 302 is being operated by remote operating device 408. It should be understood, however, that use of remote operating device 408 enables an operator of the remote operating device 408 to deploy drone 302 from vehicle 100 if the operator is located at a location remote from vehicle 100. For example, referring to FIG. 7, a control process of operating drone 302 when the operator is located remote from vehicle 100 and plans to control drone 302 using remote operating device 408 is illustrated. In the process of FIG. 7, it should be understood that operator is not located in vehicle 100 when operating drone 302.

First, in step 2000, operator may open up a software program stored on remote operating device 408 that is configured to operate drone 302. After opening the software program, a screen 510 of the remote operating device 408 can display an interface similar to that of GUI 208 (see, e.g., FIG. 6) and, the operator may input various information such as, for example, a destination for drone 302 to travel, a desired flight path, a desired altitude for the flight, a time period of the flight, and the like by interacting with various icons 500 (step 2010; FIG. 6). At this time, remote operating device 408 communicates a signal to either docking station 304 and/or ECU 402 via network 404 that drone 302 is to be deployed (step 2020) and that includes the desired information input to the remote operating device 408 by the operator.

Upon receipt of this signal and information, ECU 402 communicates with various sensors 406 and/or network 404 to determine whether conditions are acceptable to initiate deployment of drone 302 (step 2030). These conditions include obtaining information relative to current weather conditions from network 404, or obtaining signals from sensors 406 that are indicative of the weather that vehicle 100 is currently being exposed (i.e., a signal indicative of rain from rain sensors or a signal indicative of the vehicle wiper blades being operated, for example). Other conditions include whether the vehicle 100 is in motion, whether the vehicle 100 is at an orientation (i.e., on a slope) that would make it difficult to deploy drone 302 from vehicle 100, and whether panel 108 is open or removed to permit drone 302 to exit the vehicle 100 during flight. If these conditions are satisfactory, ECU 402 may transmit an instruction (step 2040) to docking station 304 to permit drone 302 to begin flight operations, detach from docking station 304, and exit vehicle 100 through opening 106 provided in roof 104 of vehicle 100. If panel 108 is obstructing opening 106, ECU 402 can instruct a motor (not shown) that moves panel 108 to move panel 108 so as to not obstruct opening 106 (i.e., controller 402 can open the sunroof, moonroof, or panoramic roof panel).

Upon receipt by docking station 304 of instructions from ECU 402, docking station 304 communicates instructions to drone 302 to initiate flight and including the desired flight characteristics input into remote operating device 408 by the operator (step 2050). Upon receipt of these instructions, drone 302 may autonomously detach from docking station 304 and exit vehicle 100 through opening 106 (step 2060). At this time, drone 302 may continue to operate autonomously using the desired flight characteristics input by the operator to remote operating device 408 before returning to docking station 304. Alternatively, drone 302 may autonomously detach from docking station 304 and exit vehicle 100 before operation of drone 302 is transferred back to remote operating device (ROD) 408 (step 2070). In such a case, drone 302 may exit vehicle 100 and reach a predetermined altitude before operation of drone 302 is transferred back to remote operating device 408. Then, the occupant can use various control icons on the display of remote operating device 408 to move in a particular direction, move at a desired velocity, reach a desired altitude, use camera 428, operate light source 430, operate depth detection device 432, and the like (step 2080). At this time, the instructions input into remote operating device 408 are communicated to ECU 402, which then communicates the instructions to docking station 304, which communicates the instructions to drone 302 via network 404.

If drone 302 is operating autonomously based on instructions input to remote operating device 408 and the desired

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flight time has elapsed, drone 302 may then automatically return to vehicle 100, pass through opening 106, and mate with docking station 304 (step 2090). Alternatively, if drone 302 is being manually operated by remote operating device 408, in step 2090 the operator can input instructions into remote operating device 408 that direct drone 302 back to vehicle 100. The operator can operate drone 302 to reenter vehicle 100 and mate with docking station 304, or the occupant can operate drone 302 to return to the operator's location exterior to vehicle 100 before manually retrieving drone 302 and manually mating drone 302 with docking station 304 when desired. It should be understood that if vehicle 100 has moved during flight operations, a location of vehicle 100 can be transmitted to drone 302 via network 404. In this regard, both drone 302 and docking station 304 may include a GPS module that may coordinate with each other.

It should be understood that during flight operations, ECU 402 may continually communicate with network 404 and/or sensors 406 to continually make a determination whether conditions are satisfactory for continuing flight operations (e.g., continually checking weather information). If for whatever reason ECU 402 determines that flight operations should cease, ECU 402 may send an instruction to docking station 304 to have drone 302 return to vehicle 100, which is then communicated to drone 302 using network 404. Moreover, it should be understood that information obtained from drone sensors 426 can be continuously transmitted from drone 302 to docking station 304 and ECU 402 via network 404. If drone sensors 426 generate signals indicative of poor operation (e.g., motor issues, poor flight conditions, and the like), ECU 402 can transmit instructions to docking station 304 that are transmitted to drone 302 via network 404 to cease flight operations and return to vehicle 100.

Now referring to FIGS. 8A and 8B, a method of surveying a potential travel path of the vehicle 100 is illustrated. At step 3000, the operator initiates a survey request. This request may be input to either remote operating device 408 or GUI 208 by selecting the appropriate icon 500 on either remote operating device 408 or GUI 208. After selecting the appropriate icon 500 to initiate the survey request, controller 436 of remote operating device 408 may communicate with GPS module 435 of remote operating device 408 to determine a location of remote operating device 408, at which time a map indicating the location of remote operating device 408 may be displayed on screen 510 of remote operating device 408 (step 3010).

At this time, the operator may select a location remote from remote operating device 408 and vehicle 100 to survey using drone 302 (step 3020) by, for example, touching the map displayed on display 510. The selected location may be, for example, an off-road trail that the operator wishes to travel using vehicle 100, or may be another location that the operator would like to survey before traveling to with vehicle 100. In the case that the selected location is an off-road trail, this is advantageous from the standpoint that the operator can view the off-road trail using camera 428 of drone 302 to determine any potential hazards or obstacles that vehicle 100 will be required to traverse while on the off-road trail. As noted in the background, this has conventionally been accomplished by first walking the off-road trail. Use of drone 302 to first survey the off-road trail can be effective in eliminating such a step.

A similar process can be used if the survey request is input to GUI 208. After selecting the appropriate icon 500 to initiate the survey request, GUI 208 may communicate with

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ECU 402 that a survey request has been input to GUI 208. If vehicle 100 is equipped with a GPS module (not shown), ECU 402 can communicate with the GPS module to receive a location of vehicle 100, and based on the signal received from GPS module indicative of the location of vehicle 100, ECU 402 can communicate this information to GUI 208 and GUI 208 can display a map of the location of vehicle 100 on display 510 of GUI 208. Alternatively, ECU 402 may receive the location of vehicle 100 from docking station 304 which is equipped with GPS module 600. Another alternative is that ECU 402 includes various off-road trail maps stored in memory 411 that may be accessed using GUI 208. In any event, at this time, the operator may select the location remote from vehicle 100 to survey using drone (step 3020) by, for example, touching the map displayed on the display 510 of GUI 208.

If the survey location has been selected using remote operating device 408, the survey location may be transmitted from remote operating device 408 to ECU 402 via network 404 (step 3030). If the survey location has been selected using GUI 208, the survey location may be transmitted from GUI 208 to ECU 402 (step 3030).

Once the survey location has been received by ECU 402, ECU 402 communicates with sensors 406 of vehicle 100 (step 3040) to confirm that the weather is suitable for flight operations (see, e.g., FIGS. 5 and 6). In addition, although not required, ECU 402 may communicate with docking station 304 to determine whether drone 302 has sufficient battery life to complete the operation. ECU 402 may also determine whether panel 108 of vehicle 100 is open. Further, if the vehicle 100 has a cargo lid 306 configured to be electronically operated by the ECU 402, the ECU 402 can determine whether lid 306 is open to permit drone 302 to deploy from docking station 304.

In step 3050, if panel 108, and optionally lid 306, are not open, ECU 402 can either communicate instructions to remote operating device 408 (via network 404) or GUI 208 to provide an indication on display 510 of these devices to the operator that panel 108 and lid 306 should be opened before flight operations can begin, or ECU 402 can send instructions to the appropriate mechanisms (not shown) that will automatically open the cargo lid 306 and panel 108 if closed.

At step 3060, the ECU 402 sends the operator-selected survey instructions to the docking station 304, which then communicates the instructions received from ECU 402 to the drone 302 (step 3070).

At step 3080, after receipt of the survey instructions from docking station 304 (including, for example, the survey location and desired flight conditions such as altitude, velocity, etc.), drone 302 can detach from the docking station 302 and exit vehicle 100 through opening 106 of vehicle 100.

At step 3090, the drone 302 executes the survey instructions selected by the operator. As noted above, the drone 302 may execute instructions to survey a specific area such as an off-road trail to assist the operator in determining whether any obstacles such as rocks, fallen trees, bodies of water, and the like or located along off-road trail that the vehicle 100 will encounter while traveling the off-road trail. Drone 302 may use camera 428 to take video footage of the surveyed area and store the footage in a memory 431 that may be viewed on display 510 once drone 302 returns to vehicle 100 and mates with docking station 304, where docking station 304 may transmit the video footage to ECU 402, which then either sends this information directly to GUI 208 or to remote operating device 408 via network 404. Alternatively, drone 302 may use camera 428 and communicate a live

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video feed to either the remote operating device **408** via network **404**, or to ECU **402** via network **404**, which then communicates the feed directly to GUI **208**. After review of the footage obtained by drone **302**, the operator may become aware of any hazards or obstacles and make a decision whether to travel the off-road trail using vehicle **100**. In addition, drone **302** may be instructed to use depth device **432** to a depth of any puddles or other bodies of water that need to be traversed by vehicle **100** along trail, which can be transmitted from drone **302** via network **404** to vehicle **100**.

At step **3100**, the operator may elect to send additional survey instructions to drone **302** or, if the survey that has been conducted by drone **302** is satisfactory to the operator, the operator may elect to recall drone **302**. For example, the operator may elect to manually control the flight and/or camera **428** of the drone **302** to further inspect other areas of interest that were found during the drone's **302** survey of the selected location (step **3110**).

At step **3120**, upon concluding the additional survey operation or recalling drone **302**, the drone **302** may navigate to the docking station **304** either autonomously or while being operated by the operator.

At step **3130**, upon reaching the vehicle **100**, the drone **302** may descend through the opening **106** and mate with the docking station **304**. Alternatively, drone **302** may be instructed to land exterior to vehicle **100** to be manually obtained by the operator and then manually mated with docking station **304**.

At step **3140**, the docking station **304** communicates to the ECU **402** that the drone **302** has mated with docking station **304** and the flight operation is concluded.

At step **3150**, the ECU **402** may then send instructions to the appropriate mechanism (not shown) to automatically close the panel **108** (and lid **306** if so equipped). Alternatively, the ECU **402** may notify the operator that panel **108** (and optionally lid **306**) should be closed by sending an instruction to remote operating device **408** (via network) or GUI **208** to display a note to the operator on display **510**.

After conclusion of operation of drone **302**, the operator may then make the decision whether or not to traverse the off-road trail using vehicle **100** based on the survey conducted by drone **302**.

The foregoing description of the embodiments has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular embodiment are generally not limited to that particular embodiment, but, where applicable, are interchangeable and can be used in a selected embodiment, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

What is claimed is:

1. A system for controlling an aerial drone, comprising:
 - a vehicle having a body defining an interior, wherein the interior includes a front row of seats, including a driver's seat and at least one passenger seat, and a cargo bin having a movable lid to provide and prevent access to the cargo bin, and wherein the cargo bin is located between the driver's seat and the at least one passenger seat;
 - a vehicle electronic control unit (ECU) located within the interior of the vehicle;

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a drone docking station positioned within the interior of the vehicle and configured to mate with the aerial drone, wherein the drone docking station is located in the cargo bin;

a docking station controller located within the drone docking station and configured to communicate with the vehicle ECU; and

at least one of a graphic user interface (GUI) provided in the interior of the vehicle and configured to communicate with the vehicle ECU and a remote operating device configured to communicate with the vehicle ECU, each of the GUI and remote operating device being configured for receipt of an input to conduct a surveillance flight using the aerial drone,

wherein upon input of a request to either the GUI or remote operating device to conduct the surveillance flight using the drone, the GUI or remote operating device is configured to transmit a signal indicative of the request to the vehicle ECU,

upon receipt of the request by the ECU, the ECU is configured to communicate an instruction to the docking station controller to initiate surveillance flight operations using the aerial drone,

upon receipt of the instruction by the docking station controller to initiate surveillance flight operations, the docking station controller is configured to communicate the instruction to the aerial drone to initiate flight and conduct the surveillance flight operation.

2. The system according to claim 1, wherein the surveillance flight operation surveys an off-road trail that is to be traversed using the vehicle.

3. The system according to claim 1, further comprising a network in communication with at least one of the vehicle ECU and docking station controller,

wherein at least one of the vehicle ECU and docking station controller is configured to communicate with the aerial drone via the network.

4. The system according to claim 3, wherein during the surveillance flight operations, the aerial drone is configured to generate at least one of images and video while in flight and transmit the images and video to at least one of the vehicle ECU and drone docking station via the network, and upon receipt of the images and video from the network, at least one of the vehicle ECU and drone docking station are configured to transmit the images and video to the GUI or remote operating device for display thereon.

5. The system according to claim 3, wherein during the surveillance flight operations, the aerial drone is configured to generate at least one of images and video while in flight and save the images and video in a memory, and after conclusion of the surveillance flight operations by the aerial drone, and the aerial drone is mated with the drone docking station, the drone docking station is configured to obtain the images and video from the memory of the aerial drone and communicate the images and video to the vehicle ECU, which then communicates the images and video to either the GUI or remote operating device for display thereon.

6. The system according to claim 3, wherein before communicating the instruction to the docking station controller to initiate surveillance flight operations using the aerial drone, the vehicle ECU is configured to communicate with at least one of the network and vehicle sensors to obtain information indicative of weather conditions that can impact the surveillance flight operation.

7. The system according to claim 6, wherein based on the information indicative of the weather conditions, the vehicle

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ECU is configured to communicate instructions to the drone docking station that either permit the surveillance flight operations to occur or prevent the surveillance flight operations from occurring.

8. A method of using an aerial drone to conduct surveillance of an off-road trail before travelling the off-road trail using an off-road vehicle, the off-road vehicle including a vehicle body having an interior, wherein the interior includes a front row of seats, including a driver's seat and at least one passenger seat, and a cargo bin having a movable lid that opens and closes, comprising:

receiving, by either a remote operating device or a graphic user interface (GUI) of the off-road vehicle, a request to conduct a surveillance flight of the off-road trail using the aerial drone;

retrieving, by either the remote operating device or GUI and from a vehicle electronic control unit (ECU) having a memory, a digital map of the off-road trail stored in the vehicle ECU's memory;

displaying on a screen of either the remote operating device or GUI the digital map of the off-road trail;

selecting a location of interest along the displayed digital map of the off-road trail;

transmitting, by either the remote operating device or GUI, a request to conduct a surveillance flight at the selected location of interest to the vehicle ECU;

determining, by the vehicle ECU, an opened/closed state of the movable lid of the cargo bin;

generating, by the vehicle ECU, an instruction to conduct a surveillance flight at the selected location of interest along the off-road trail;

opening, by the vehicle ECU, the movable lid of the cargo bin if the determined opened/closed state of the movable lid is closed;

transmitting, by the vehicle ECU, the generated instruction to conduct a surveillance flight at the selected location of interest to a docking station controller; and transmitting, by the docking station controller, the generated instruction to conduct a surveillance flight at the selected location of interest to the aerial drone.

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9. The method according to claim 8, wherein the off-road vehicle includes an interior that includes a drone docking station that is configured to mate with the aerial drone, the drone docking station including the drone docking station controller that communicates with each of the aerial drone and the vehicle ECU.

10. The method according to claim 8, wherein the off-road vehicle includes a plurality of sensors in communication with the vehicle ECU that are configured to generate signals indicative of weather conditions being experienced by the off-road vehicle, and

before generating the instruction to conduct surveillance, the vehicle ECU is configured to determine, based on the signals indicative of the weather received from the plurality of sensors, whether the weather conditions are satisfactory for conducting the surveillance of the off-road trail.

11. The method according to claim 10, wherein the vehicle ECU is configured to communicate with a network and obtain information indicative of the weather conditions from the network, and

before generating the instruction to conduct surveillance, the vehicle ECU is configured to determine, based on the information indicative of the weather received from the network, whether the weather conditions are satisfactory for conducting the surveillance of the off-road trail.

12. The method according to claim 8, wherein the vehicle ECU and aerial drone are each in communication with a network, and during the surveillance of the off-road trail using the aerial drone, the aerial drone is configured to transmit images of the off-road trail to the vehicle ECU via the network, and

wherein the method further comprises transmitting the images received by the vehicle ECU to either the remote operating device or GUI for viewing on the screen.

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